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## **Route Selection and Obstacle Avoidance with a Short-Range Haptic Sensory Substitution Device**

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### **Abstract**

The design of Sensory Substitution Devices (SSDs) often relies on the belief that the information supplied by the devices should allow the construction of spatial mental representations on the basis of which routes are planned. This study, in contrast, illustrates that navigation using an SSD can be conceived as an on-line, dynamic process, without the need for establishing a predefined plan or model of the task prior to its execution. We analyzed route selection performed with a vibrotactile SSD that could detect environmental surfaces only within a short spatial range, limiting the availability of information about remote parts of the environment to be navigated. Sixty sighted participants performed a navigation task that involved the goal of reaching a target destination while avoiding five obstacles (placed in randomly predetermined configurations). Three groups of participants differed in the sensory modality used (restricted visual, acoustic + vibrotactile, and restricted visual + vibrotactile). While participants in the visual condition had fewer obstacle collisions and reached the target location sooner, the groups coincided to a large extent in terms of the routes that they followed. Furthermore, the routes selected by participants in all groups conformed well to routes predicted by a dynamic model of visually-guided locomotion (Fajen and Warren, 2003). These findings show that local, limited information about environmental layout can support route selection equivalent to that seen when information about the full environmental layout is available.

**Keywords:** enactive torch, information-based model, route selection, sensory substitution, vibrotactile stimulation, visual impairment

## 1. Introduction

Route selection and obstacle avoidance are essential tasks for autonomous navigation. Visually impaired and blind individuals are known to experience problems with their autonomy due to difficulties with avoiding obstacles and walking toward targets. Sensory Substitution Devices (SSDs) have been designed to help people cope with the absence of vision, using either touch or hearing as substituting senses. SSDs that are designed to help people navigate autonomously in the absence of vision are typically classified as special-purpose SSDs (Loomis et al., 2012) because, rather than substituting wholesale for vision, they aim to provide a substitute for vision in specific situations—in this case, enhancing mobility in space. Several of these devices, which are also known as electronic travel aids, are based on information about the distance to objects measured with lasers, ultrasonic signals, or infrared sensors (Dakopoulos and Bourbakis, 2010; Liu et al., 2010).

The development of effective sensory substitution technology for autonomous navigation stands to benefit from a theoretical guiding framework (Lenay et al., 2003; Lobo et al., 2018). There are two major approaches that aim to explain navigation behavior of blind individuals in terms of their mental representations of the environment (Thinus-Blanc and Gaunet, 1997). The first approach, known as the cumulative or deficiency model (Kolarik et al., 2017; Schinazi et al., 2016), states that representations are key to navigation behavior and that these representation are visual in nature. According to this approach, congenitally blind and early-blind individuals should show considerably worse spatial performance than late-blind and blindfolded-sighted individuals. Evidence that is consistent with this view has indeed been reported (Bigelow, 1991; Eimer, 2004; Millar, 1976; Pasqualotto and Nevell, 2007; Pasqualotto et al., 2013). The second representation-based approach holds that the relevant representations are amodal. This leads to the expectation that the differences between individuals with different blindness onset times should be limited. This second approach has also received empirical evidence (Tinti et al., 2006). The evidence partly comes from studies conducted with SSDs (e.g., Chebat et al., 2011; Stronks et al., 2015).

Notwithstanding the debate concerning the onset of blindness and performance, the visual representation and amodal representation approaches share the belief that navigation behavior is based on internal representations. An alternative approach, referred to as the information-based control approach, holds that representations are not necessary because navigation is controlled on-line using information that is detected as the individual's activity unfolds (Kolarik et al., 2017). The predictions of this approach regarding differences between early-blind and late-blind individuals are similar to those of the amodal representation approach. The representational and on-line control approaches differ, however, with regard to the claimed role of route planning. Explicit, a priori route planning is indispensable in the representational approaches and irrelevant in the information-based control approach. This difference affects the design of SSDs. According to the representational approaches, SSDs should provide a sufficiently complete description of the scene to make route planning possible. In contrast, according to the information-based control approach, the user should have access to information sufficient for on-line control and for making perceptual-motor decisions as needed in the moment, meaning that the SSD does not need to provide the user with a complete description of the layout.

More generally, advancing the design of special-purpose SSDs for navigation requires identification of the information sufficient for effective navigation (Lobo et al., 2014; Lobo et al., 2018). As long as this information is provided to users with sufficient precision, it seems reasonable to assume that the lesser the complexity of the SSD, the better. For many tasks, however, it remains to be determined what information users need, and, consequently, how minimalistic the assistive SSD can be. Relatedly, Faugloire and Lejeune (2014) have argued that there is a shortage of studies on tactile SSDs and navigation that report quantitative measures of performance other than task duration. Early studies by Jansson (1983), Guarniero (1977), and Zelek et al. (2003) were optimistic about the possibilities of tactile SSDs for autonomous navigation, but these studies did not report detailed measures of performance. The lack of such measures makes comparisons of the efficacy of SSDs and of the utility of the information they provide difficult.

Our aim in this study, then, was twofold. First, we aimed to test whether a particular minimalist SSD—the Enactive Torch (ET; Froese et al., 2012), consisting of a single vibrotactile actuator attached to the wrist—is sufficient to provide the information relevant to accurately perform a complex navigation task involving the selection of a particular locomotor route when several such routes are possible. This differs from the maze navigation task explored in Froese et al., which was a task constrained largely by avoiding collisions with nearby walls. Second, we aimed to illustrate that route selection with a vibrotactile SSD like the ET in a real-world environment may occur in an on-line manner, as claimed by the information-based control approach, rather than requiring complete, a priori visual or amodal representations.

The intensity of the vibrotactile stimulation provided by the ET is dependent on the distance to a surface detected by a laser range finder. The stimulation provided by the ET is also contingent on the users' exploratory movements—on how the user scans the environment for nearby surfaces—which has been argued to be a key aspect of following a unique pathway in a maze (McGann et al., 2011; cf. Díaz et al., 2012; Froese et al., 2012; Lobo et al., 2018). An important feature of the ET with regard to our second aim, in particular, is that it provides information about the egocentric distance in a single direction over a relatively short spatial range (between 20 and 150 cm). This means that no direct information is available concerning allocentric distances and that, at a certain moment and from a certain position, only a part of the whole spatial layout can be detected. For users of the ET it is therefore impossible to represent and pre-plan a full route from the beginning of the task. It follows that the routes of users of the ET necessarily emerge during the action and are shaped by the information users detect as they traverse the environment. Hence, if route selection with the ET matches route selection under visual guidance, this would offer tentative support for the information-based approach to route selection, which, for visually-impaired navigation, is the least explored explanation for route selection.

The most prominent information-based model to study route selection and obstacle avoidance is the one proposed by Fajen and Warren (2003) and Fajen et al. (2003). This model has

been used, for example, to study the interception of moving targets (Fajen and Warren, 2007), robot navigation (Huang et al., 2006), and collective behavior in crowd locomotor dynamics (Bonneaud et al., 2012). In the model, obstacles are described as repellers and targets (or goal locations) as attractors. The strength of the repellers and attractors depends on their distance to the point of observation. The model is consistent with the view that route selection is an emergent behavior that does not need a predefined plan that is first set up and then executed.

More formally, Fajen and Warren’s (2003) model describes locomotion toward a target with a differential equation that is based on the person’s direction of motion,  $\phi$ :

$$\ddot{\phi} = -b\dot{\phi} - k_g(\phi - \psi_g)(e^{-c_1 d_g} + c_2) + \sum_{i=1}^{\#obstacles} k_o(\phi - \psi_{o_i})e^{-c_3|\phi - \psi_{o_i}|}(e^{-c_4 d_{o_i}}), \quad (1)$$

with  $\ddot{\phi}$  the angular acceleration,  $\dot{\phi}$  the angular velocity,  $\psi_g$  the goal angle (with respect to the agent, in an egocentric framework),  $\psi_{o_i}$  the angle of each obstacle,  $d_g$  the goal distance, and  $d_{o_i}$  the distance to each obstacle. The remaining terms  $b$ ,  $k_g$ ,  $k_o$ ,  $c_1$ ,  $c_2$ ,  $c_3$ , and  $c_4$  are parameters that affect variables with regard to damping, stiffness, and degree of attraction and repulsion. To the best of our knowledge, despite the relevance of this dynamic model, it has not been applied in sensory substitution research, and only one previous sensory-substitution study used a related experimental paradigm (Lobo et al., 2018).

We performed an experiment with a navigation task in which route selection was essential to avoiding obstacles and reaching a target. The experiment included three between-group conditions in which participants received visual, auditory, and/or vibrotactile information using the ET. In the first condition, participants wore vision-reducing goggles simulating a severe loss of vision with similarities to tunnel vision, a symptom common in eye diseases such as glaucoma or retinitis pigmentosa (Robinson et al., 1997; Vargas-Martin and Peli, 2006). In this condition—referred to as V—the length of the detectable visual field was not restricted, so participants could detect the target

and all obstacles from the start, albeit with a small field of view that required visual scanning. In the second condition, referred to as S+ET condition, a sound source (S) was placed near the target and blindfolded participants used the ET for obstacle detection. In this condition the target could be detected auditorily from the beginning of the trial, but, due to the limited range of the ET, the obstacles could not. The third condition, referred to as V+ET, was an intermediate condition in which participants wore the vision-reducing goggles (V) and used the ET.

Regarding the first aim of this study, one may expect that obstacle detection with (even restricted) vision would be superior to obstacle perception with the ET. Because of this, we hypothesized that better performance would be observed in the V and the V+ET condition than in the S+ET condition in terms of trial duration (i.e., durations would be shorter) and the number of times that obstacles were touched (i.e., fewer obstacles would be touched). However, we believed that the ET can provide information relevant to complete a navigation task. Thus, we hypothesized that participants in the S+ET condition would reach the target on the same proportion of trials as participants in the V and the V+ET conditions.

Regarding our second aim, the fact that the range in which obstacles can be detected in the S+ET condition is shorter than in the V and the V+ET conditions means that it is impossible in the S+ET condition to plan the full route to the target in advance. Consistent with the information-based control theory, however, we expected that such route planning would not be necessary. As a consequence, we hypothesized that recorded locomotor paths would conform to the predictions of Fajen and Warren's (2003) information-based model in each modality condition.

## **2. Method**

### **2.1. Participants**

Sixty students of the University of Cincinnati (36 women and 24 men,  $M_{age} = 20.6$  years,  $SD = 2.8$  years) participated in the experiment. None of the participants had used sensory substitution or similar assistive devices before. All participants were right-handed and had normal vision or corrected-to-normal vision with contact lenses. None of them reported mobility impairments that

could affect their performance. Participants received course credit in return for their participation. The research project was approved by the University of Cincinnati Institutional Review Board (IRB). Written informed consent was obtained from all participants.

## **2.2. Apparatus**

A schematic representation of the experimental set-up can be seen in Figure 1. The exploration area was a  $500 \times 700$  cm open area in a laboratory room with a carpeted floor. Five rectangular foam obstacles of  $90 \times 27 \times 21$  cm (height, width, depth) were placed in the area in configurations described below. A cylindrical target with a diameter of 9 cm and a height of 210 cm was placed at the center of the farthest edge of the area. An eight-camera motion capture system (Optotrak Certus, Northern Digital Inc., Canada) registered the position of active motion-capture markers located on the top of the head (attached to a cap) and one marker on the right wrist. The position of the markers was registered at 120 Hz.

The vibrotactile device used in the experiment was the Enactive Torch v. 5 (Figure 2a; Favela et al., 2014; Favela et al., 2018; Froese et al., 2012). This battery-powered device consists of an infrared sensor that works as a rangefinder and a vibrotactile actuator that is attached to the wrist with Velcro band. The infrared sensor covers a range of 20 to 150 cm in a straight line that coincides with the pointing direction. An Arduino pro-mini microcontroller converts the distance to the first-encountered object, as measured by the sensor, into a voltage level that is transmitted to the actuator. The intensity of vibration of the actuator is inversely proportional to the distance measured by the sensor. That is, when the distance to the object increases the intensity of vibration decreases and, conversely, when the distance to the object decreases the intensity of vibration increases. The voltage level delivered to the actuator is maximally 5 V at a distance of 20 cm and minimally 0 V at a distance of 150 cm.

In the S+ET condition only, participants heard pink noise at 80 dB intensity emitted by a speaker placed just behind the target at a height of 125 cm. In the other two conditions, V and V+ET, participants wore vision-reducing goggles fashioned from welding goggles. The goggles

consisted of an opaque surface with a hole of a diameter of 0.2 cm (approximately  $3^\circ$  of visual angle; Figures 2b and 2c), simulating impaired vision.

### **2.3. Design**

The primary experimental manipulation concerned the modalities available to guide performance. Participants in the first condition, the V condition, performed the task with the goggles, but without the ET and without any sound. Participants in the second condition, the S+ET condition, performed the experiment while pink noise was played at the target location, blindfolded, and using the ET. Participants in the third condition, the V+ET condition, wore the vision-reducing goggles and used the ET.

Ten obstacle layouts were selected following three stages: First, we created a vector of  $x$  coordinates from -2.5 to 2.5 m in steps of 0.5 m and another vector for  $y$  coordinates from 0 to 6 m also in steps of 0.5 m. Second, we randomly selected five numbers of each vector to create ten layouts with five obstacles. Third, simulations following Fajen and Warren's (2003) model were used to discard configurations that led to trivial or impossible trajectories. The first 10 layouts that fulfilled these criteria were used in the experiment. The  $x$  and  $y$  coordinates of the obstacle locations in each layout are shown in Table 1. These ten layouts were randomly divided in two series of five. Each participant experienced only one series of five obstacle configurations; half of the participants received the first series, and half received the second series. Each participant performed 20 trials, which included four repetitions of the same series of obstacle configurations presented in random order within each series. After a random sequence of 20 trials was created for each series, that random order was presented to half of the participants and was then reversed for the other half of participants.

### **2.4. Procedure**

Prior to the experiment, participants were provided with a brief explanation of the task and the apparatus. They received the following instructions: "You will begin at the starting position in one end of the room and asked to walk toward a target on the other side. On your way, you may

encounter foam obstacles that you should try to avoid. Your task consists of touching the target with your left hand without touching the obstacles.” The experimenter showed participants one of the obstacles, the target (but not its position), and the starting position, mentioning that the location of the obstacles changed between trials. The use of foam obstacles and a carpeted floor, along with a careful placement of the obstacle by the experimenter, minimized audio cues coming from the change of obstacle positions. A brief condition-specific practice session was performed before the 20 experimental trials.

#### *2.4.1. Practice Trials*

*V practice.* Participants, standing near the target, first turned so as to stand with their back facing the target and put the goggles on. The target was then moved to the left or right and the participants were instructed to turn and visually search for the target. After this single-trial practice, the experimenter accompanied the participants to the starting point, where they waited with their back toward the experimental area.

*S+ET practice.* Participants in the S+ET condition were first assessed on their ability to localize the sound. While blindfolded, they were placed at two different locations, and asked to point to the location of the source of the pink noise (i.e., the target location). Participants were then given the opportunity to explore the functioning of the ET, pointing to one hand while holding the device with the other hand. The experimenter then placed the target and one obstacle behind the participants and gave them the instruction to turn around and inspect the two objects with the ET, reporting their respective heights and widths to assure differentiation between the target and an obstacle. After that, participants turned 180° and the target was moved backward. The experimenter placed an obstacle 250 cm from the participant in the direction of the target and 50 cm displaced to the right side. Participants were instructed to turn and to walk toward the target while scanning on their right side with the ET. The practice session ended when participants reported that the obstacle was located immediately to their right.

*V+ET practice.* Participants in this condition first received the same short practice as participants in the V condition. After that, they completed the same practice with the ET as participants in the S+ET condition.

#### 2.4.2. *Experimental Trials*

After the condition-specific practice session, the experimenter placed the motion-capture markers on the participant's body. Participants moved to the starting position and waited with their back toward the experimental area. The experimental trials started with the experimenter saying "go" in the V and V+ET conditions, and with the onset of the pink noise in the S+ET condition. At the beginning of the trials participants turned around and began walking to the target. A trial finished when the participant touched the target with their left hand, after which the experimenter accompanied the participant back to the starting position along a direct route (after another experimenter had removed the obstacles). Between trials, the experimenter rearranged the spatial configuration of the obstacles.

#### 2.5. *Data Processing*

Performance errors and locomotor paths were analyzed. The experimenter manually registered the performance errors, defined as touches of the obstacles. One trial (0.1% of total) was not registered and it was considered missing data in further analyses concerning performance errors. Locomotor paths were recorded with the motion capture system and participants' trajectories were registered using the head markers. In total, 39 trials (3.3%) had irremediable data recording errors and those trials were discarded in further analyses that involved data recorded with the motion capture system. For the rest of the trials ( $n = 1161$ ), a mean of 14.21% of frames were not properly registered and those data were interpolated with the function *interp1* in Matlab (Mathworks, Inc.). The data were filtered with a forward and backward 4<sup>th</sup>-order low-pass Butterworth filter with a cutoff frequency of 0.6 Hz. Similarly to the data processing in Fajen and Warren (2003), in order to compare the observed trajectories to the trajectories predicted by the model, we binned each value of the participants' trajectories in the  $x$  coordinate (see Figure 1 for detail about the axes) in

intervals of 10 cm along the  $y$  axis. Usually, the observed trajectories ended before the target due to the distance between the top of the head, where the tracking markers were placed, and the extended arm that touched the target. Each trial started with participants' backs facing the target and a subsequent rotation of approximately  $180^\circ$ . For these reasons, when comparing observed and predicted trajectories, we discarded the first and the last meter of the binned data along the  $y$  axis. The starting position for each modeled trajectory was the first value from its corresponding observed trajectory.

### 3. Results

#### 3.1. Performance Variables

Participants successfully reached the target in 100% of the trials. They completed the task without touching any obstacle (a performance error) in 69.7% of the trials, with one error in 22.0% of the trials, and with two or more errors in 8.3% of trials (Table 2). Participants needed, on average, 29.3 s ( $SD = 16.2$ ) to complete the trials.

To compare the performance errors and trial durations among the modality conditions, we averaged out the negligible effects of trial order and spatial configuration. This was done by averaging the measures over subsets of four participants that used different trial orders and different series of obstacle configurations. A main advantage of this procedure was that it allowed us to maintain the normality assumption in the following ANOVA. Without averaging, the trials without errors, and hence zeros in the data, would have led to violation of the normality assumption.

Performance errors were analyzed with a one-way between-subjects ANOVA with modality (V, S+ET, and V+ET) as the factor. A significant main effect was observed,  $F(2, 12) = 6.3$ ,  $p = .013$ ,  $\eta_p^2 = .51$ . Subsequent post-hoc analyses using Tukey's HSD tests ( $\alpha$  level = .05) revealed only that there were significantly fewer errors in the V condition than in the S+ET condition (Table 3).

A second one-way, between-subjects ANOVA with modality as the factor was performed with trial duration as dependent variable. There were significant differences among the conditions,

$F(2, 12) = 12.9, p = .001, \eta_p^2 = .68$ . Tukey's HSD tests revealed significant differences between the V condition and both the V+ET and S+ET conditions (Table 3). In the latter conditions, the mean duration per trial was approximately 1.6 and 2 times longer, respectively, compared to the V condition.

### 3.2. Locomotor Paths

The locomotor paths were analyzed with regard to the spatial configuration of obstacles and the modality condition (Figure 3). For each observed trajectory, we computed the associated predicted trajectory using the dynamic model defined in Equation 1. We used the parameter values of simulations #1 and #2 of Fajen et al. (2003):  $b = 3.25$ ,  $k_g = 7.5$ ,  $k_o = 198$ ,  $c_1 = 0.4$ ,  $c_2 = .4$ ,  $c_3 = 6.5$ , and  $c_4 = 0.8$ . The simulations of Fajen et al. assumed a constant velocity of 1 m/s, which we reduced to 0.25 m/s so as to match our participants' velocity as estimated based on the observed average trial durations. The distances between observed and predicted trajectories for each binned data point (difference  $d_i$  in Figure 4) were averaged in each trial. The mean distance between the observed and predicted trajectories ( $\text{Distance}_{OP}$ ) for all trajectories was 38.7 cm ( $SD = 29.1$ ; Table 3). We conducted a one-way, between-subjects ANOVA to test the effect of modality (V, V+ET, and S+ET) on  $\text{Distance}_{OP}$ . The ANOVA did not reveal a significant effect of modality,  $F(2, 1158) = 2.4, p = .09$ .

The routes followed by participants were analyzed per spatial configuration of obstacles. A route was defined as a distinct path around the obstacles and terminating at the target that was followed for more than four trajectories per spatial configuration and per modality condition. For example, in Figure 5, route A (which is the inner route between routes B and C) includes trajectories that went to the right of the first, fourth, and fifth obstacles and to the left of the second and third obstacles. In one spatial configuration of obstacles (#8) only one route was identified, in six spatial configurations (#2, #3, #4, #5, #6, #10) two routes were identified, and in three spatial configurations (#1, #7, #9) three routes were identified. In total, 22 routes were identified and 973 trajectories (83.8% of all observed trajectories) followed one of these routes. We studied the

corresponding predicted trajectories of these 973 observed trajectories and we found that in 66.0% of trials ( $n = 642$ ) the route of the predicted trajectory was the same as the route of the observed trajectory. Moreover, in 84.7% of the trials ( $n = 824$ ) the predicted trajectory was included in one of the observed routes. There was high similarity of the routes chosen in each modality condition. The V+ET condition shared 81.0% of the identified routes with both the S+ET and the V conditions, while the V condition shared 81.8 % of the routes with the S+ET condition, as can be noted in Figure 6 where all grouped trajectories in a route were averaged in each modality condition.

The absolute distance between each trajectory and its corresponding averaged route was calculated as a measure of the dispersion of the trajectories ( $\text{Dispersion}_{\text{OA}}$ ). On average, the  $\text{Dispersion}_{\text{OA}}$  within a route was 20.7 cm ( $SD = 6.4$ ). This dispersion measure was analyzed with a one-way ANOVA with modality (V, V+ET, S+ET) as the between-subjects factor. There was a significant effect of modality,  $F(2, 55) = 6.4, p = .003, \eta_p^2 = .19$ . Tukey HSD tests revealed two significant differences between modalities (Table 3): The V condition had a lower dispersion compared to both the V+ET and the S+ET conditions.

#### 4. Discussion

The first aim of this experiment was to test if the information provided by the ET would be sufficient to solve a complex navigation task. Participants reached the target on 100% of trials, regardless of modality condition. No trials were dismissed due to participants feeling lost in the exploration area or being unable to avoid the obstacles. In the S+ET condition 61.4% of the trials did not show any performance error (i.e., touches of obstacles). A brief comparison of our results with previous studies suggests that the ET helps users to avoid obstacles equally well or even better than other SSDs. For example, Maidenbaum et al. (2014) reported that participants collided once per trial during the last trial of a task performed inside a corridor, Kolarik et al. (2017) reported 93% successful trials in a task in which participants had to detect and circumvent only one obstacle, and Chebat et al. (2011) reported a percentage of correct responses slightly below 60% for blindfolded, sighted individuals who used an electro-tactile device applied to the tongue to avoid obstacles. This

suggests the ET, at least when supplemented with a sound source at the target location, sufficed for solving the navigation task.

As hypothesized, however, better performance was observed in the V condition than in the S+ET condition in terms of the number of errors and trial duration. In addition, trajectories included in the identified routes showed a larger dispersion around the average trajectories in the S+ET condition than in the V condition. Thus, while the ET allowed for an adequate level of ultimate task performance, the details of participants' navigation behavior were by no means equivalent across modality conditions. It is possible that with additional practice, navigational behavior with the ET device—which was entirely novel to our participants—could improve relative to what we observed (Chebat et al., 2015; Lobo et al., 2014). Nonetheless, it appears there is room to improve the utility of the ET as a navigational aid for negotiating complex environments.

Participants in the V+ET condition used reduced vision as well as the ET. Their results were more similar to the S+ET condition than to the V condition. In particular, they showed worse performance than participants in the V condition in terms of trial duration and the dispersion of their trajectories. This may seem surprising because participants in the V+ET condition could have relied on reduced vision on its own (as in the V condition) by ignoring the ET. In our interpretation, this unexpected finding shows that the ET, rather than being perceived as annoying and to be avoided, attracted the attention of participants, who explored and used the device despite the availability of reduced vision. Although these results indicate a lack of added benefit of the ET when there is residual vision, the evoked attention may be an advantage for blind or nearly blind users in the first stages of training with a vibrotactile SSD. This is so because the motivation to explore the possibilities offered by the SSD, even in the absence of immediate performance benefits, may be a necessary condition for an efficient use of the SSD on the longer term. Also, a possible explanation for the lack of added benefit of the ET when vision was available could be fatigue related to head movements. It is possible that participants tried to reduce exploratory movements performed with

the head that caused fatigue by using their wrist and arm to explore more with the ET. The differences in exploratory movements among conditions should be addressed in future research.

The second aim of this study was to explore if route selection with a vibrotactile SSD may occur in an on-line manner, as predicted by the information-based control approach. As hypothesized, the rate of coincidence among the three modality conditions with regard to the routes that were followed was higher than 80%. Our analyses therefore did not reveal substantial differences between the average spatial trajectories of routes followed by ET users and those followed by participants with reduced vision. Moreover, we did not observe significant differences between modality conditions with respect to the distance between the observed trajectories and the trajectories predicted by a model with parameter values for a visually guided task (Fajen and Warren, 2003; Fajen et al., 2003). Given that participants in the S+ET condition had access to nearby obstacles but not distant ones, these findings are consistent with the theory that route selection does not necessarily imply planning the full route in advance on the basis of visual or amodal spatial representations. As noted, the dispersion of trajectories was lowest in the V condition. From an information-based perspective, this finding could indicate a difference in the information available in (and thus the actions afforded by) the different modality conditions.

The experiment did not include a condition in which participants utilized only the ET. Our preliminary testing revealed that it took participants a considerably longer amount of time to complete the task in this case. We therefore coupled the use of the ET with the sound (S+ET) or reduced vision (V+ET). While it is possible that using the sound source to indicate the location of the target could have provided participants with a very crude sense of the spatial layout of the task environment prior to the beginning of a trial, it was the configuration of obstacles more than the location of the goal that mattered for shaping the particular locomotor paths that were observed. Any (somewhat coarse) representation of the target's location would therefore not seem to be sufficiently rich to serve as a basis for locomotor route selection to avoid the obstacles. Therefore,

the results of the S+ET condition can be interpreted as support for the information-based control approach.

Nonetheless, the necessity of including a sound source could provide reasons for diminished optimism about currently available minimalist SSDs. Additional research is needed to indicate how minimalist devices can optimally distinguish targets and obstacles. In addition, although we have demonstrated that the navigation task can be performed with a minimalist SSD, our experiment does not rule out possible benefits of a larger number of sensors and actuators and of a longer spatial range of detection. Previous studies have shown that a higher number of sensors and actuators enhances the acuity of SSD-based perception (Lobo et al., 2018). In the present task, a higher number of sensors and actuators might have led to fewer performance errors and less dispersion of the observed trajectories.

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**Table 1***Coordinates of the Obstacles in the Ten Spatial Configurations*

| Configuration | Coordinates (x, y) of the Five Obstacles |       |       |       |       |       |       |       |       |       |
|---------------|------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Number        | $x_1$                                    | $y_1$ | $x_2$ | $y_2$ | $x_3$ | $y_3$ | $x_4$ | $y_4$ | $x_5$ | $y_5$ |
| 1             | -0.5                                     | 3     | 1.5   | 3     | -1    | 3.5   | 0.5   | 4     | 0     | 5     |
| 2             | 0.5                                      | 1.5   | -1    | 3.5   | -1.5  | 4     | 0     | 4     | -1    | 5.5   |
| 3             | 0.5                                      | 2     | 0.5   | 2.5   | -1    | 4     | -0.5  | 4     | -1.5  | 4.5   |
| 4             | 1.5                                      | 2     | -0.5  | 2.5   | 0.5   | 2.5   | -0.5  | 3.5   | -1    | 5     |
| 5             | -0.5                                     | 2     | 1.5   | 3.5   | 0     | 3.5   | -1    | 3.5   | -0.5  | 6     |
| 6             | 0.5                                      | 3     | -1    | 3     | 1     | 4     | 0     | 4.5   | 0     | 6     |
| 7             | 0                                        | 1.5   | 1     | 1.5   | 1.5   | 5     | 0     | 5.5   | -0.5  | 5.5   |
| 8             | 0.5                                      | 1.5   | 1.5   | 2.5   | -1.5  | 2.5   | -0.5  | 4.5   | -1    | 4.5   |
| 9             | 0                                        | 1.5   | 1     | 2     | -1.5  | 2     | 1.5   | 3.5   | 0     | 5.5   |
| 10            | 0.5                                      | 2     | 1     | 3.5   | -0.5  | 3.5   | 1.5   | 4.5   | 0.5   | 6     |

*Note.* Values are given in m from the starting position (coordinates:  $x = 0$ ,  $y = 0$ ).

**Table 2***Number and Percentage of Performance Errors for Each Modality Condition*

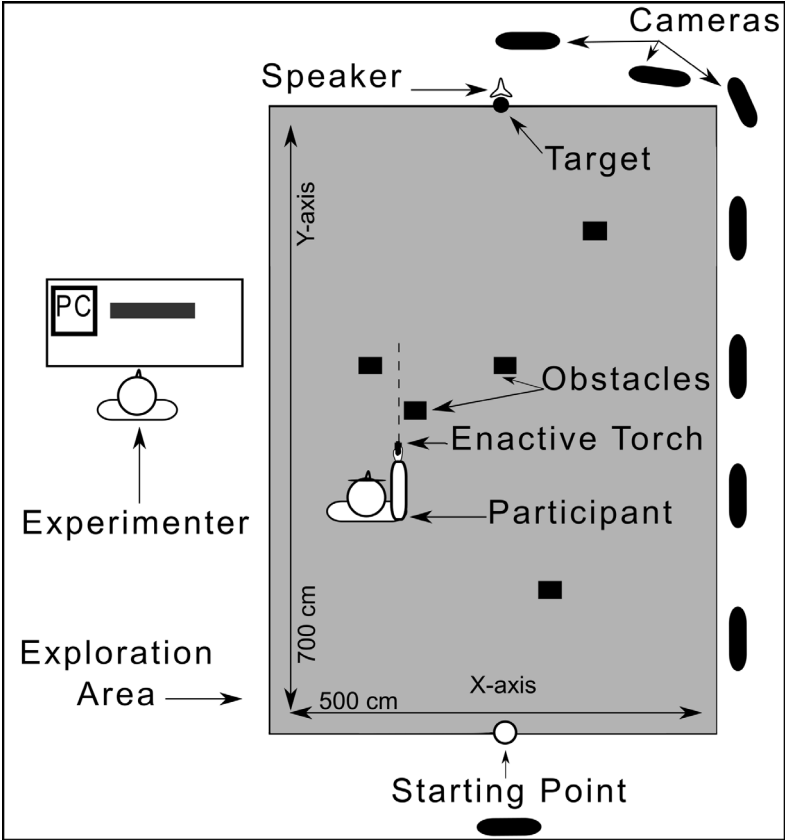
| Performance Errors | Modality Condition |             |             |        |
|--------------------|--------------------|-------------|-------------|--------|
|                    | V                  | S+ET        | V+ET        | Total  |
| None               | 326 [81.5%]        | 245 [61.4%] | 265 [66.3%] | 69.7 % |
| One                | 58 [14.5%]         | 109 [27.3%] | 97 [24.3%]  | 22.0 % |
| Two                | 16 [4.0%]          | 37 [9.3%]   | 33 [8.3%]   | 7.2 %  |
| Three              | 0 [0.0%]           | 8 [2.0%]    | 4 [1.0%]    | 1.0 %  |
| Four               | 0 [0.0%]           | 0 [0.0%]    | 1 [0.25%]   | 0.1 %  |

*Note.* Percentage data between brackets refer to percentage of performance errors within the modality conditions.

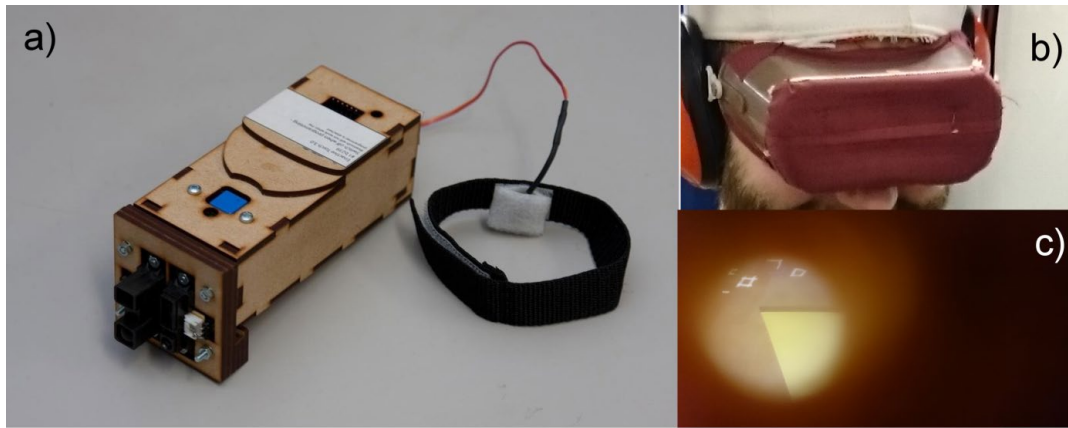
**Table 3***Means and SDs for Performance and Movement Variables in each Modality Condition*

| Modality<br>Condition | Number of Errors<br>Per Trial | Trial Duration          | Distance <sub>OP</sub>    | Dispersion <sub>OA</sub> |
|-----------------------|-------------------------------|-------------------------|---------------------------|--------------------------|
| V                     | 0.2 <sub>a</sub> (0.1)        | 19.4 <sub>a</sub> (2.6) | 36.3 <sub>a</sub> (29.7)  | 16.9 <sub>a</sub> (5.8)  |
| S+ET                  | 0.5 <sub>b</sub> (0.2)        | 38.1 <sub>b</sub> (5.9) | 40.9 <sub>a</sub> (30.1)  | 23.1 <sub>b</sub> (5.9)  |
| V+ET                  | 0.4 <sub>a,b</sub> (0.2)      | 30.9 <sub>b</sub> (7.8) | 38.8 <sub>a</sub> (27.3). | 22.1 <sub>b</sub> (6.0)  |
| Total                 | 0.4 (0.2)                     | 29.5 (9.6)              | 38.7 (29.1)               | 20.7 (6.4)               |

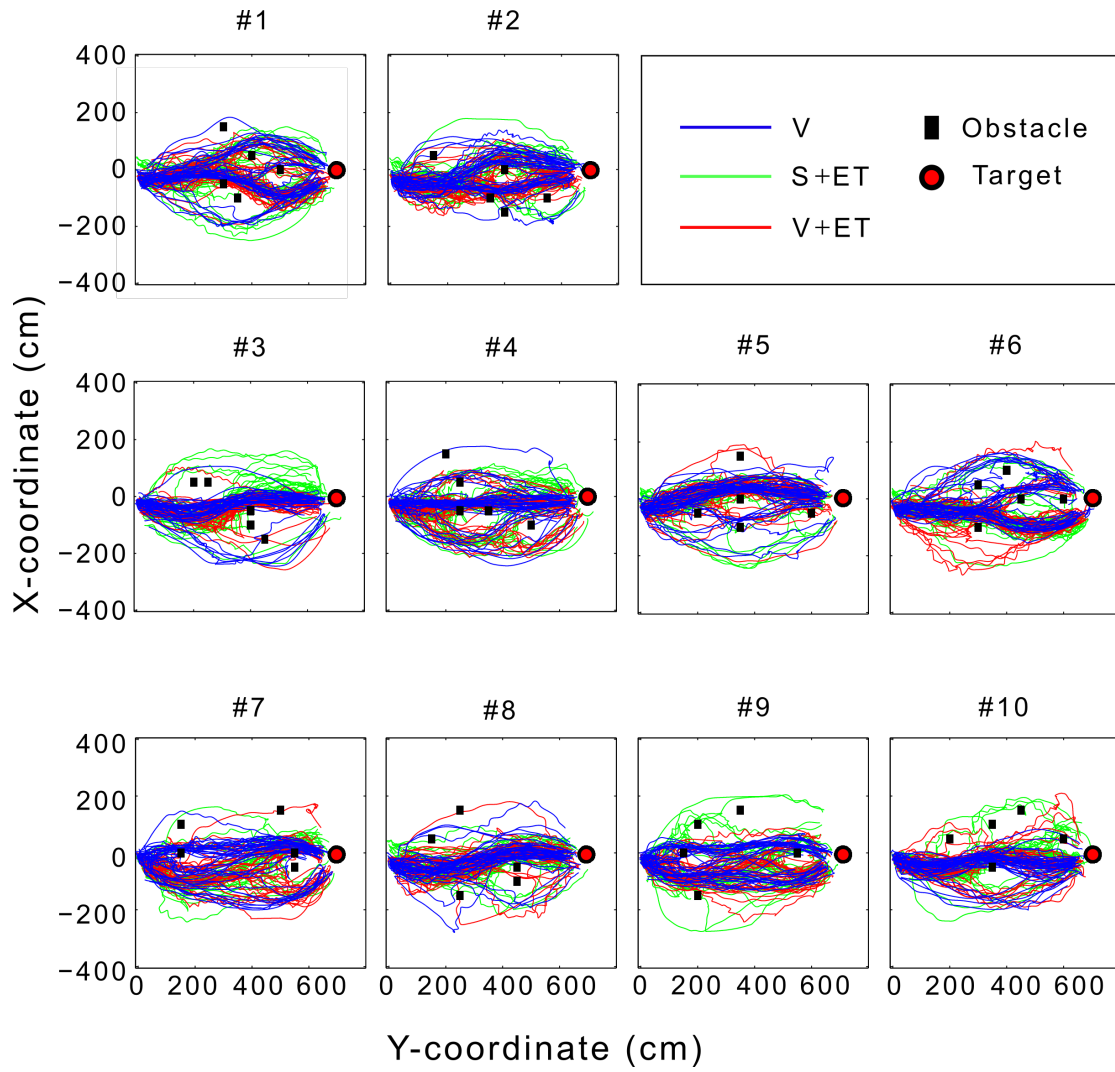
*Note.* Each subscript letter denotes a subset of variable modality condition whose observed means do not differ significantly from each other at the .05 level.



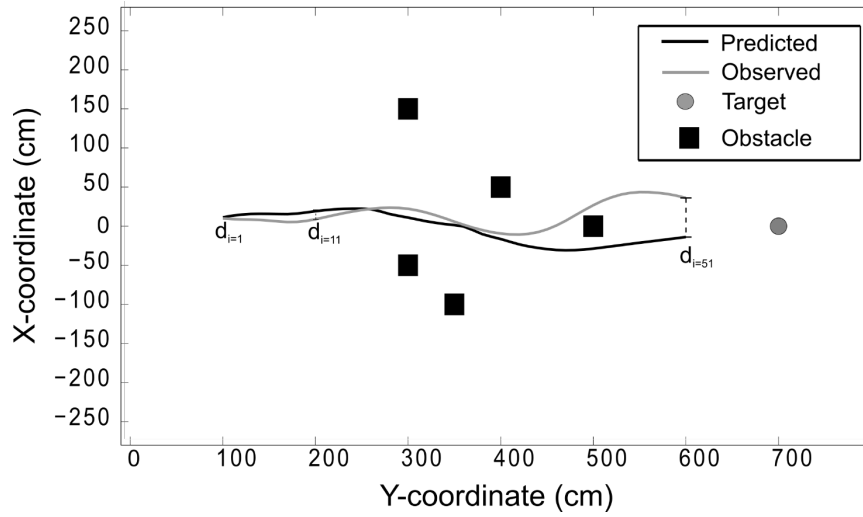
**Figure 1. Experimental set-up.** Top view of experimental set-up with exploration area.



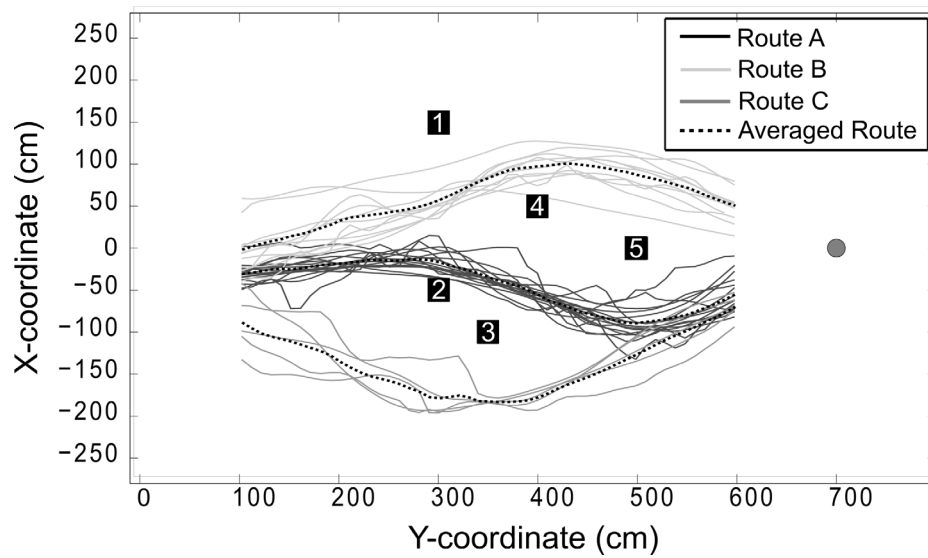
**Figure 2. Experimental apparatus.** (a) Enactive Torch v. 5 (picture retrieved from <http://enactivetorch.files.wordpress.com/2008/01/et5.jpg>, July 8, 2016). (b) Vision-reducing goggles. (c) Picture taken through the vision-reducing goggles.



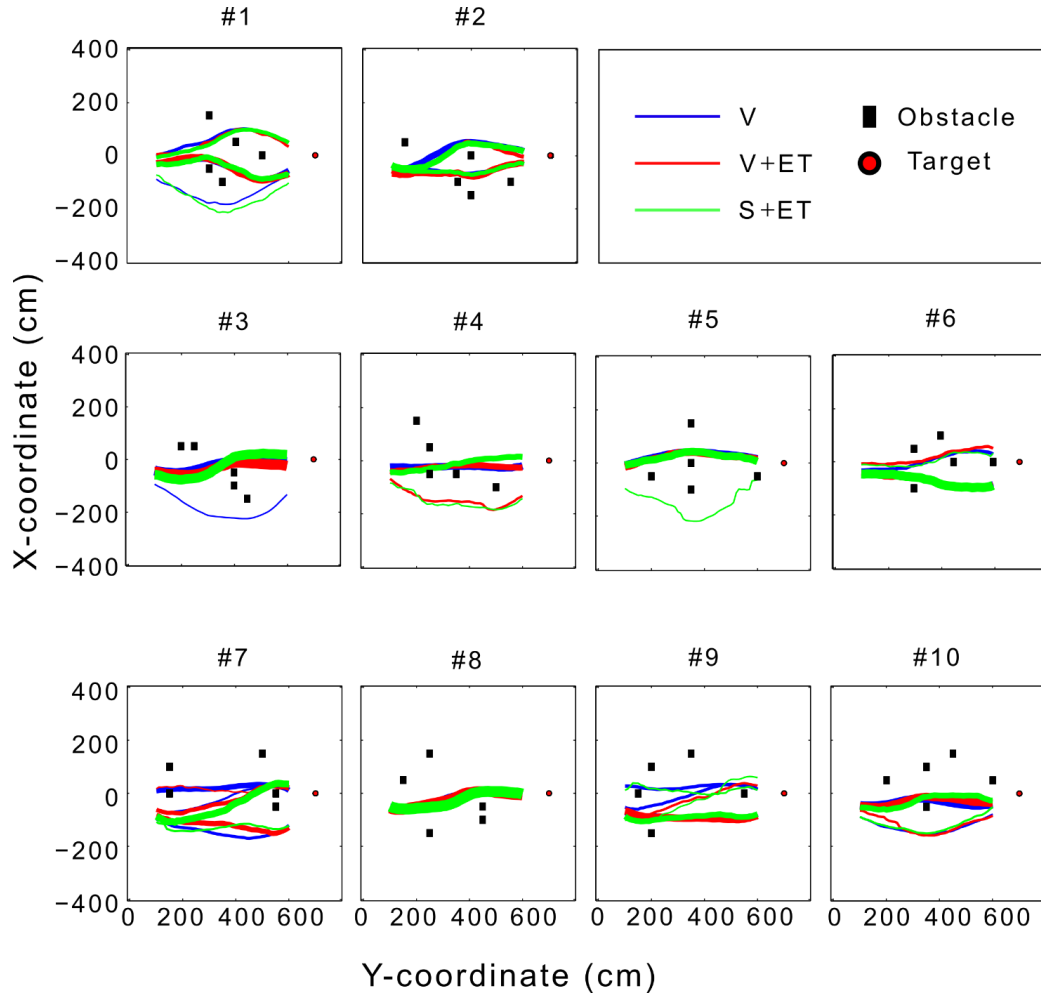
**Figure 3. Recorded participant trajectories.** The number above each of the ten subplots indicates the specific spatial configuration as shown in Table 1. Each subplot includes 10 Participants  $\times$  3 Conditions  $\times$  4 Repetitions = 120 trajectories. The target is enlarged regarding the scale of the figure for viewing purposes.



**Figure 4. Single-trial example of predicted and observed trajectories.** The distance between the observed and predicted trajectories in the  $x$  direction is the average absolute value of  $d_i$  calculated every 10 cm along the  $y$  axis; in this example it was 22.0 cm. The predicted trajectory was computed from the first  $x$  coordinate of the observed trajectory, implying that  $d_{i=1} = 0$ .



**Figure 5. Example of route identification.** The three routes shown in the figure (A, B, and C) were the routes that we identified with the  $10 \text{ Participants} \times 4 \text{ Repetitions} = 40$  trajectories for the V condition and for spatial configuration #1. The dotted lines represent the average of all trajectories included in one route.



**Figure 6. Average trajectories for each identified route.** The average trajectories are shown per spatial configuration and per modality condition. The width of the colored lines is a function of the number of trajectories that are included in the route. For example, in spatial configuration #1, the thin blue line is the average of 5 trajectories (the minimum number of trajectories to produce a route); in spatial configuration #8, the thick green line is the average of 36 trajectories.

## Author Biographies



Lorena Lobo, Ph.D., is adjunct lecturer (docente nivel IV) at the Universidad a Distancia de Madrid. Her experimental work is focused on sensory substitution and augmented reality for navigation tasks. Other interests include post-cognitivist approaches in perceptual learning, the perception of affordances with different modalities, and the benefits of applying an embodied approach to new technological developments. Her theoretical work deals with historical aspects related to the development of the ecological account in psychology.



Patric Nordbeck is in the Ph.D. program in the Department of Psychology at the University of Cincinnati. His primary research area involves mathematical modeling and simulation of nonlinear complex system dynamics, drawing from ecological psychology, dynamic systems theory, and chaos theory. He is currently creating interdisciplinary research projects using nonlinear methods and statistics in neuropsychology and industrial organization.



Vicente Raja is postdoctoral researcher at the Western University Canada. His main field of research is philosophy of cognitive science, paying special attention to embodied approaches to perception, action, and cognition. Other philosophical interests include history and philosophy of science, philosophy of technology, and aesthetics. He also develops experimental work in the field of plant signaling and behavior.



Anthony Chemero, Ph.D., is Professor of Philosophy and Psychology at the University of Cincinnati. His research is both philosophical and empirical; typically, it tries to be both at the same time. His research is focused on questions related to nonlinear dynamical modeling, ecological psychology, complex systems, phenomenology, and artificial life. He is the author of *Radical Embodied Cognitive Science* (2009, MIT Press) and, with Stephan Käufer, *Phenomenology* (2015, Polity Press).



Michael A. Riley, Ph.D., is Professor of Psychology and Director of the Center for Cognition, Action, and Perception at the University of Cincinnati. His research investigates sensorimotor control with emphases on motor variability and perception-action coupling using methods inspired by complex dynamical systems theory and with applications focusing on improving motor deficits in sports medicine and related fields.



David M. Jacobs, Ph.D., is Associate Professor (Profesor Contratado Doctor) at the Department of Psychology of the Universidad Autónoma de Madrid. His main theoretical contributions are formulated around an ecological theory of learning that he, with Claire F. Michaels, has named direct learning. His most recent empirical research aims to apply and advance ecological ideas about perception, action, and learning, in the fields of sensory substitution, aviation, and sports.



David Travieso, Ph.D., is Associate Professor (Profesor Contratado Doctor) and Director of the Perception and Action Research Group at the Universidad Autónoma de Madrid. His research interests are haptics and sensory substitution, as well as the theory and history of psychology. His approach emphasizes the perception of affordances and perception-action coupling, classic concepts in the ecological approach.